

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation(s) that you entered into the calculator and the result.

1. The volume of a shipping box in the shape of a right rectangular prism is 382.5 cubic inches. The height of the prism is 12 inches (in.) and the length of the prism is 7.5 in.

a. Determine the width, in inches, of the shipping box.

work	
Width	4.25 inches

- b. The shipping box is made of cardboard and used to ship paper. Complete the statement.

The amount of cardboard used to construct the shipping box can be determined

by finding the

- ☐ volume
☒ surface area

of the box, while the amount of paper the

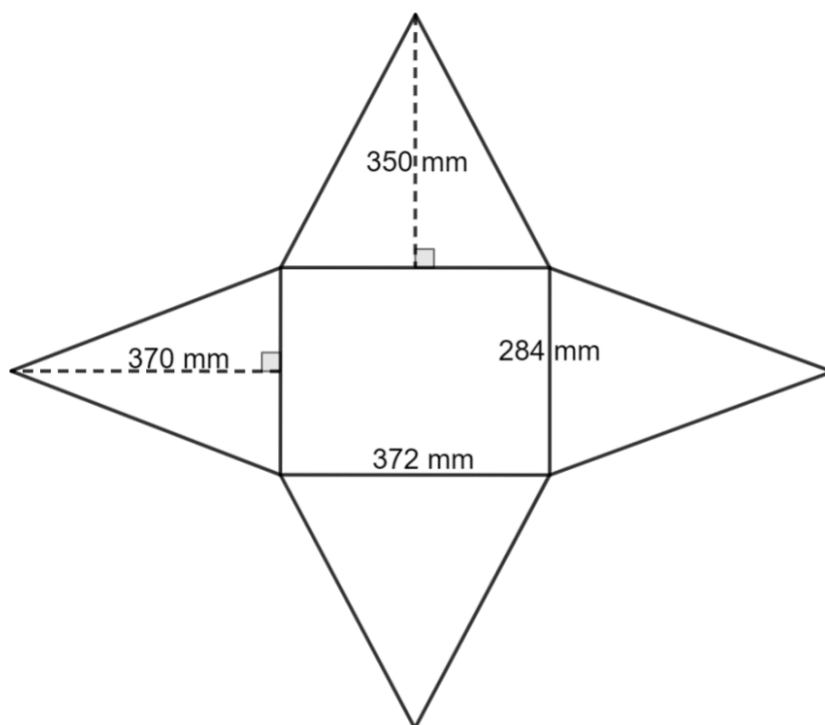
shipping box can hold can be determined by finding the

- ☒ volume
☐ surface area

of the box.

2. The net of a right rectangular pyramid with dimensions in millimeters (mm) is shown.

Determine the surface area, in square millimeters, of the right rectangular pyramid.



work

Surface Area

340,928 square millimeters

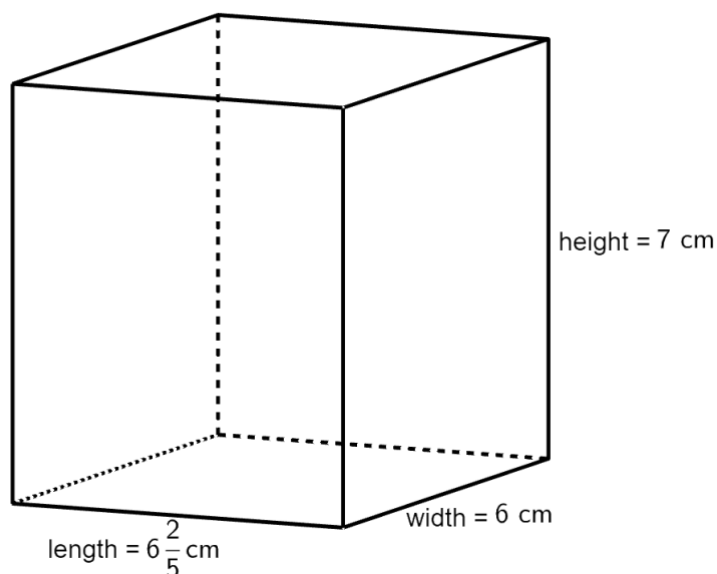
3. A rectangular prism with dimensions in centimeters (cm) is shown.

a. Complete the statements.

To find the volume of the prism,

cubes with side lengths of

- ☐ $\frac{1}{2}$ cm
- ☒ $\frac{1}{5}$ cm
- ☐ $\frac{1}{6}$ cm
- ☐ $\frac{1}{7}$ cm



can be used so that the length will have 32 cubes, the width will have

30 cubes, and the height will have 35 cubes. This results in

33,600 cubes, each with a volume of

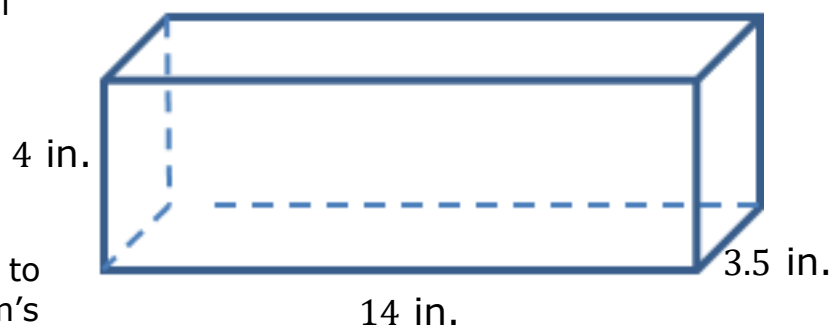
- ☐ $\frac{1}{8}$
- ☒ $\frac{1}{125}$
- ☐ $\frac{1}{216}$
- ☐ $\frac{1}{343}$

cubic centimeters.

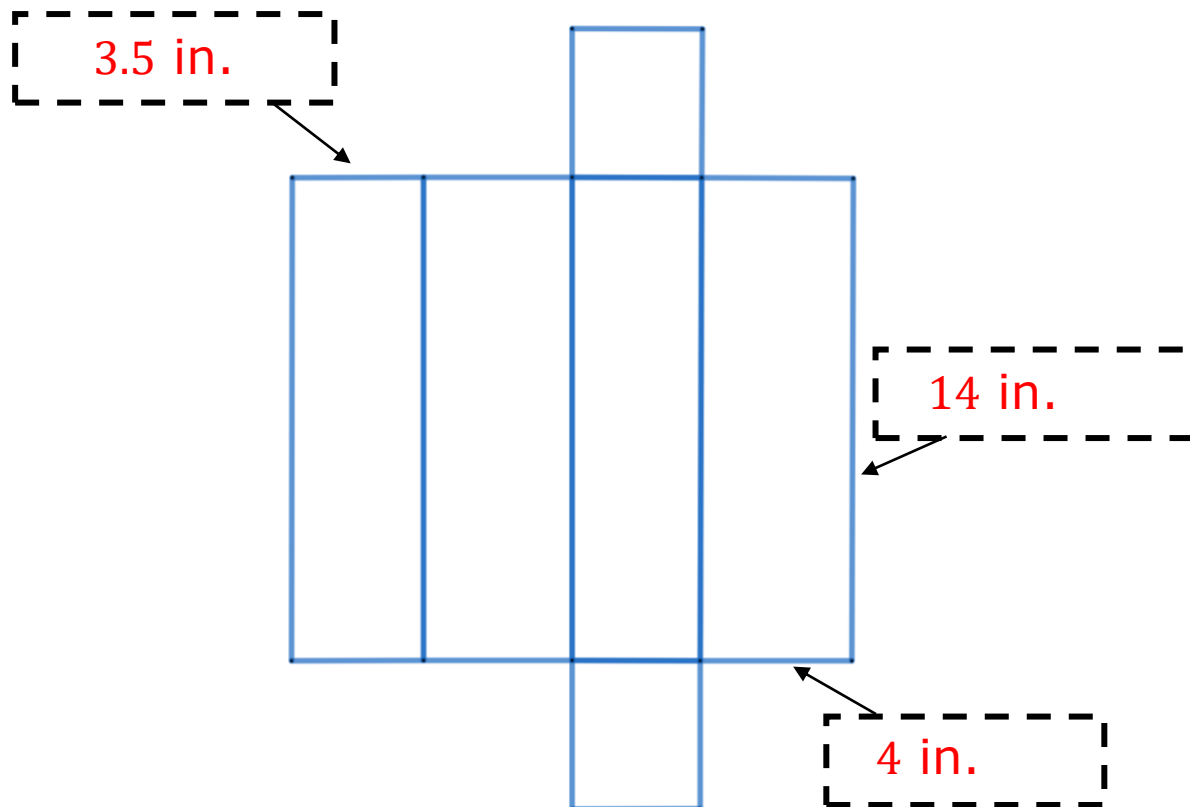
b. Determine the volume, in cubic centimeters, of the right rectangular prism.

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Volume	$268\frac{4}{5}$ or 268.8 cubic centimeters

4. A rectangular prism is shown with a length of 14 inches (in.), a width of 3.5 in., and a height of 4 in.



- a. Use the dimensions of the prism to label the dimensions of the prism's net. Include the units.



- b. Determine the surface area of the prism.

work

Surface Area	238 square inches
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